6.8.1 Description of boxplot

(i) If the median is near the center of the box, the distribution is approximately symmetric

(ii) If the median falls to the left of the center of the box, the distribution is positively skewed.

(iii) If the median falls to the right of the center of the box, the distribution is negatively skewed.

(iv) If the lines are about the same length, the distribution is approximately symmetric

(v) If the right line is larger than the left line. the distribution is positively skewed.

(vi) If the left line is larger than the right line. the distribution is negatively skewed.

Remark:

(i) The line drawn from minimum value of the dataset to Q_1 and Q_3 to the maximum value of the data set is called whisker.

(ii) Box plot is also called Box – Whisker plot.

(iii) A box and whisker plot illustrate the spread of the distribution and also gives an idea of the shape of the distribution

Example 6.20

The following data gives the Number of students studying in XI standard in 10 different schools 89,47,164,296,30,215,138,78,48, 39 construct a boxplot for the data.

Solution:

Step 1: Arrange the data in order

30,39,47,48,78,89,138,164,215,296

Step 2: Find the median

$$\text{Median} = \frac{78+89}{2} = 83.5$$

Step 3: Find Q_1

30,39,47,48,78

$Q_1 = 47$

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